

Project Report: Supply Prescriptive Analytics

Name: Geetha

Project Title: Supply Prescriptive Analytics

Objective

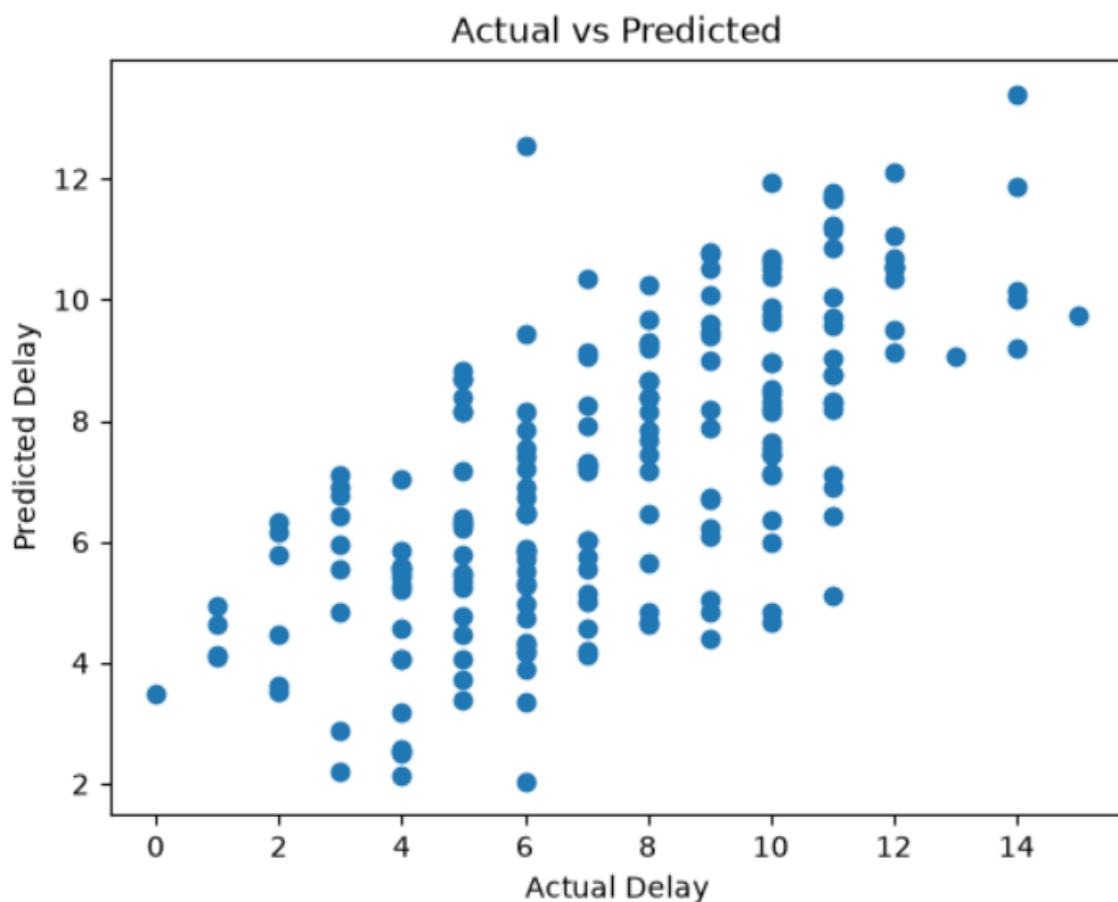
The objective of this project is to predict shipment delays using machine learning and provide optimal decisions to handle delays using prescriptive analytics.

Week 1: Predictive Analytics

In Week 1, I built a predictive model using the XGBoost algorithm to predict shipment delays. The model was trained on sample supply chain data and generated predictions based on input features.

To evaluate the model, I compared the actual delay values with predicted delay values using a graph. This helped in understanding the model performance and accuracy.

OUTPUT:



Week 2: Prescriptive Analytics

In Week 2, I implemented prescriptive analytics using optimization techniques. When a delay is predicted, the system generates three strategies:

- **Low Cost Strategy:** Lower cost but slower delivery
- **Fast Delivery Strategy:** Faster delivery but higher cost
- **Balanced Strategy:** Moderate cost and delivery time

Each strategy includes actions, total cost, time impact, and trade-offs. This helps in making better decisions based on the situation.

OUTPUT:

⚠ Delay Predicted!

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Strategy: Low Cost

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Actions: ['Air Shipping', 'Extra Inventory', 'New Supplier']

Total Cost: \$1600.00

Time Impact: 10.00 days

Trade-off: Cheaper but slower

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Strategy: Fast Delivery

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Actions: ['Extra Inventory']

Total Cost: \$300.00

Time Impact: 5.00 days

Trade-off: Faster but expensive

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Strategy: Balanced

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Actions: ['Extra Inventory', 'New Supplier']

Total Cost: \$800.00

Time Impact: 8.00 days

Trade-off: Balanced cost and speed

Tools & Technologies Used

- Python
 - XGBoost
 - SciPy
 - Jupyter Notebook
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Conclusion

This project helped me understand how predictive and prescriptive analytics work together. The model predicts delays, and the optimization suggests the best possible actions, improving decision-making in supply chain management.
